



DPUTimer

Implements a extended stop watch.

1. Inputs:

<i>Start</i>	When switching from 0 to 1 the stop watch is started or resumed.
<i>Stop</i>	When switching from 0 to 1 the stop watch is stopped.
<i>Reset</i>	When switching from 0 to 1 the saved time of the stop watched is reset to 0.
<i>Timer</i>	Predefined duration, which shall be compared to the elapsed time (milliseconds)

2. Outputs:

<i>RunTime</i>	Elapsed time (milliseconds)
<i>StopTime</i>	Stopped time (milliseconds)
<i>Greater</i>	=1, if $RunTime > Timer$ (else 0)
<i>Less</i>	=1, if $RunTime < Timer$ (else 0)
<i>Equal</i>	=1, if $RunTime = Timer$ (else 0)
<i>GreaterEqual</i>	=1, if $RunTime \geq Timer$ (else 0)
<i>LessEqual</i>	=1, if $RunTime \leq Timer$ (else 0)

3. Functional principle:

As soon as the *Start* input is set, the timer starts to run and the *RunTime* starts to count. At *Stop* a timer is stopped, after a new *Start* it continues to start running from the previous saved time.

Via the *Reset* input the saved time can be set back to 0.

The elapsed time (*RunTime*) is compared to a predefined *Timer* value in every simulation step. The compared outputs (*Greater*, *Less*, *Equal*, *GreaterEqual* and *LessEqual*) are set appropriate.